

Adaptive Compressed Residual: Redundancy-Aware Mechanism for Compact Multimedia Representation Learning

Meng'en Qin, Yu Song, Jing Fang, Deyu Meng, *Member, IEEE*, Xiaohui Yang, and Lili Yang

Abstract—Residual connections are a fundamental component of deep neural networks and have been widely adopted in various architectures. Despite their success, conventional element-wise residual addition indiscriminately propagates both informative and redundant activations, potentially limiting representation compactness and generalization, especially in deep layers. In this work, we revisit residual learning from a redundancy-aware perspective and propose Adaptive Compressed Residual (ACR), a learnable enhancement to conventional residual connections. Instead of directly injecting identity features, ACR performs learnable sparse compression on the skip pathway before fusion. The compression process is implemented via an unfolded sparse-coding module inspired by the Iterative Shrinkage-Thresholding Algorithm (ISTA), enabling selective suppression of task-irrelevant components while preserving discriminative information. To adapt to varying network depths and noise conditions, the compression strength is learned end-to-end. We further provide theoretical characterization under classical sparse recovery assumptions to offer insight into the adaptive behavior of the proposed mechanism. Extensive experiments on image, audio, and video classification, together with object detection and semantic segmentation, demonstrate the applicability of ACR across diverse multimedia tasks and backbone architectures.

Index Terms—Residual connections, Adaptive information compression, Unfolding model, Representation learning

I. INTRODUCTION

RESIDUAL connections have become a fundamental building block in modern multimedia representation learning, enabling the successful training of deep architectures across a wide range of modalities, including images, videos, audio, and multimodal signals. Since their introduction in convolutional neural networks (CNNs), residual pathways have been extensively adopted in representative architectures such as ResNet [1], MobileNet [2]–[4], and ConvNeXt [5]. By allowing features to bypass intermediate transformations,

residual connections effectively stabilize optimization, alleviate vanishing gradients, and facilitate the construction of very deep networks.

Beyond CNNs, residual connections play an equally critical role in Transformer-based [6] and Mamba-style [7] sequence models. In both language and vision Transformers, residual addition is used to combine the outputs of self-attention or feed-forward modules with their inputs, ensuring stable information flow across layers. Similar designs are also observed in emerging sequence models for audio and temporal data. This simple element-wise fusion has proven effective in preserving low-level information while enabling higher-level semantic abstraction, making residual connections a de facto standard across diverse multimedia architectures [8].

However, the most straightforward implementation of a residual connection—namely, the direct element-wise sum of input and transformed features—may not always yield the best possible expressivity or information flow. This naive addition treats all channels and spatial locations equally, ignoring potential differences in feature importance or compatibility. Motivated by this observation, recent studies have explored enhanced residual designs, including attention-based modulation, gating mechanisms, and adaptive weighting strategies. While these methods introduce flexibility into residual fusion, they primarily focus on reweighting residual signals rather than explicitly removing redundant information. Moreover, most existing approaches [9]–[11] lack a principled mechanism to control the accumulation of redundant features, and their effectiveness is often validated empirically without theoretical characterization.

Empirical analysis reveals that both CNN-based and Transformer-based residual networks exhibit substantial redundancy in deep layers. We visualize class activation heatmaps of two representative architectures: ResNet (CNN-based) and ViT (Transformer-based) in Fig. 1. It can be observed that ResNet exhibits significant redundancy in features before the classifier, paying attention to many areas unrelated to the classification target. Due to its self-attention mechanism, ViT exhibits more concentrated activation patterns than ResNet, although redundant responses are still observed. From an information-theoretic perspective, the above issue can be interpreted through the lens of the Information Bottleneck (IB) principle, which posits that deep representations should progressively discard input redundancy while preserving task-relevant information. Empirical and theoretical studies suggest that shallow layers primarily capture detailed input characteris-

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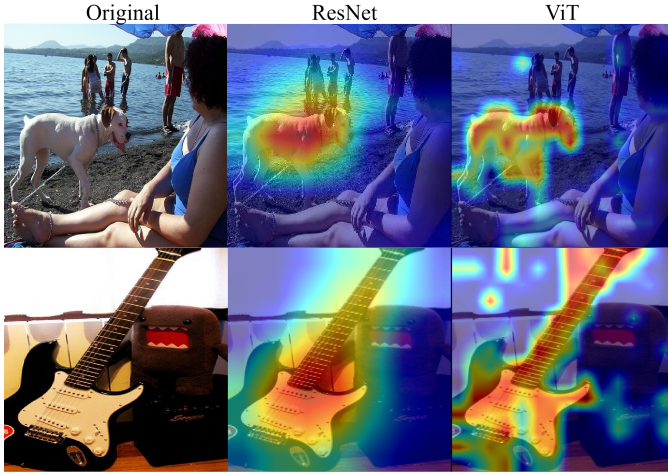


Fig. 1: Visualization of class activation heatmaps generated by Grad-CAM for ResNet and ViT.

tics, whereas deeper layers are expected to perform abstraction and compression [12]. However, conventional residual connections indiscriminately preserve input features across all depths, thereby counteracting the desired compression behavior in deep networks. This mismatch between residual design and information-theoretic objectives motivates the need for an explicit mechanism that regulates information flow in skip pathways.

In this work, we propose Adaptive Compressed Residual (ACR), a redundancy-aware residual mechanism that explicitly compresses skip-path features before fusion. Instead of directly adding the input feature to the transformed output, ACR adopts a compress-then-inject strategy: residual features are first selectively compressed to retain task-salient components and then merged with the main branch. This design enables the network to suppress task-irrelevant activations while preserving the optimization benefits of residual learning. To implement this idea in a principled and efficient manner, we formulate residual compression as a sparse coding problem that jointly accounts for reconstruction fidelity, sparsity, and task relevance. The resulting formulation is efficiently solved by unfolding a fixed number of iterations of the Iterative Shrinkage-Thresholding Algorithm (ISTA) into a lightweight neural module, enabling seamless integration into existing CNN, Transformer and Mamba based architectures.

To accommodate varying noise levels and network depths, ACR further introduces a learnable adaptive compression coefficient that is optimized end-to-end with the task loss. This allows each residual module to automatically adjust its compression strength without manual tuning. From a theoretical perspective, we establish support-recovery and prediction-error guarantees under standard Restricted Eigenvalue conditions. Extensive experiments on image, video, and audio benchmarks demonstrate that ACR reduces residual redundancy, improves feature compactness, and enhances generalization across diverse architectures, including CNNs, Transformers, and Mamba-based models. Rather than introducing another task-specific architectural tweak, this work revisits residual

learning from a redundancy-aware perspective and provides a unified mechanism for improving representation efficiency.

The main contributions of this paper are:

1. We introduce a learnable information compression module into every residual link via a “compress-then-inject” strategy. By reformulating feature merging as a Lasso optimization with reconstruction, sparsity, and task-loss terms—and unfolding a fixed number of ISTA iterations into convolutional/deconvolutional layers—ACR efficiently filters out redundant activations and passes task-salient information forward with minimal overhead.

2. We propose an implicit task-loss term $\mathcal{L}_{task}(\tilde{X}, Y)$, derived from classification cross-entropy, which variationally maximizes a lower bound on the mutual information $I(\tilde{X}; Y)$ without explicit estimation. This loss drives the compressed features to retain only those components most predictive of the downstream objective and integrates seamlessly into end-to-end training.

3. To avoid manual tuning, we endow each residual module with a learnable compression coefficient that automatically adjusts sparsity levels across varying noise conditions and network depths. We provide support-recovery and prediction-error bounds under standard Restricted Eigenvalue conditions, and demonstrate on multiple CNN, Transformer and Mamba backbones that ACR consistently reduces deep-layer redundancy and boosts model accuracy.

II. RELATED WORK

A. Improved Residual Architectures

Beyond the original residual architectures in ResNet, numerous studies have explored enhanced residual formulations to improve representational capacity and training dynamics. SENet [13] introduces adaptive channel reweighting within residual blocks, while SKNet [14] and ResNeSt [15] further improve multi-branch feature aggregation and attention-guided fusion. Residual Attention Networks [9] extend this idea by stacking attention modules to refine residual representations hierarchically. CBAM [16] sequentially applies channel- and spatial-attention mechanisms to multiplicatively recalibrate intermediate feature representations. In addition, gating and dynamic routing mechanisms have been proposed to regulate information flow through shortcut pathways. Residual Gates [17] introduce learnable scalar control of residual contributions, while more recent cross-layer interaction schemes such as DeepCrossAttention [18] enable input-dependent aggregation of features across layers. These methods regulate feature propagation primarily through channel or spatial reweighting, gating, branch aggregation, or dimensionality reduction. In contrast, ACR operates directly on the shortcut representation and formulates its compression as a sparse-reconstruction problem before residual fusion. Therefore, ACR differs from these approaches by explicitly combining a reconstruction constraint, sparsity regularization, and task-guided supervision to control the information propagated through the shortcut, rather than relying only on multiplicative feature recalibration or scalar gating.

Optimization-inspired architectures provide another relevant line of research. LISTA [19] unfolds an iterative shrinkage-thresholding algorithm into a fixed-depth trainable network to approximate sparse-code inference, whereas ReDuNet [20] derives an entire interpretable network architecture from the principle of rate reduction. These methods demonstrate how optimization procedures can be translated into trainable neural architectures. In contrast, ACR does not replace or redesign the complete backbone. Instead, it embeds a fixed-depth unfolded sparse-reconstruction module locally into the shortcut pathway while retaining the original transformed branch. Accordingly, the distinction of ACR does not lie in introducing a new general-purpose unfolding algorithm, but in using task-guided unfolding as a plug-in mechanism to realize “compress-then-inject” residual fusion, together with a learnable block-specific regularization parameter that adapts the compression strength across network depths.

B. Information Bottleneck

The information bottleneck theory [21], [22] provides an elegant balance between data compression and information preservation from the perspective of information theory. This theory suggests that the ideal feature representation should form an intermediate representation T by compressing the input data X , so that T minimizes mutual information with X (compression redundancy) while maximizing mutual information with the target task Y (preserving discriminative information). Tishby et al. [23] conducted an in-depth analysis of neural networks using the theoretical framework of this principle, and calculated the optimal information-theoretic limit of Deep Neural Networks (DNNs) using this representation method. At the same time, a new optimization criterion for DNNs based on information theory was proposed. Kawaguchi et al. [24] provided strict generalization boundaries in mathematical theory, demonstrating why this control over compression and preservation in information bottlenecks is desirable in deep learning. Despite these insights, most research remains theoretical, with challenges still to be addressed for effective integration with practical deep learning methods.

C. Multimodal Representation Learning

Learning unified and transferable representations across heterogeneous modalities has become a central topic in modern multimedia research. Recent progress in large-scale vision-language pretraining, such as CLIP [25] and ALIGN [26] demonstrates that shared embedding spaces learned from cross-modal supervision can significantly improve generalization and downstream task performance. Beyond pairwise modality alignment, a line of research focuses on building unified architectures capable of processing multiple modalities within a single backbone. Representative examples include Omnivore [27], which jointly models images, videos, and audio streams, and Uni-Perceiver [28], which introduces a shared representation space for diverse sensory signals. Similar trends are also observed in general-purpose multimodal foundation models [29]–[31], highlighting the importance of modality-invariant feature abstraction for scalable multimedia understanding.

III. METHODOLOGY

In this section, we analyze the redundancy introduced by element-wise addition in residual connections. To address this issue, we formulate an information-compressed residual objective consisting of reconstruction, sparsity, and task-supervision terms. The sparse-reconstruction component is implemented by unfolding a fixed number of ISTA iterations. The task-supervision loss is then computed, and all model parameters are jointly optimized through a single backward pass. Building on this formulation, the compression strength is made learnable, allowing the model to adapt to different feature statistics and network depths. The resulting adaptive design is further analyzed theoretically in terms of its parameter-estimation error. Fig. 2 illustrates the overall framework of the proposed method.

A. Information Redundancy in Residual Connections

Residual connections are widely adopted in deep architectures for stabilizing optimization. In a standard residual block, the output is

$$\tilde{Y} = F(X) + X, \quad (1)$$

where X is the input feature and $F(X)$ denotes a nonlinear transformation. While this additive design facilitates identity mapping and mitigates vanishing gradients, it also directly propagates all components of X , regardless of their relevance to the downstream task. As depth increases, such unfiltered reuse may lead to accumulated redundancy and reduced feature discriminability.

To conceptually analyze this effect, we consider the feature associated with each sample as an element of an abstract d -dimensional feature space and decompose this space into two orthogonal subspaces:

$$\mathbb{R}^d = V \oplus U, \quad (2)$$

where V represents task-relevant components correlated with the target Y , and U represents task-irrelevant components. This decomposition is used only as an analytical abstraction and does not assume explicit knowledge of V or U in the actual implementation. Under Eq. (1), the identity shortcut does not explicitly suppress components in U , allowing task-irrelevant information to persist across layers.

For clarity, we consider the following linearized residual network:

$$X_{\ell+1} = X_{\ell} + W_{\ell}X_{\ell}, \quad (3)$$

where $X_{\ell} \in \mathbb{R}^d$ denotes the abstract feature representation at layer ℓ , and $W_{\ell} \in \mathbb{R}^{d \times d}$ denotes the corresponding linearized transformation. According to the above decomposition, W_{ℓ} can be partitioned as

$$W_{\ell} = \begin{bmatrix} W_{\ell}^{VV} & W_{\ell}^{VU} \\ W_{\ell}^{UV} & W_{\ell}^{UU} \end{bmatrix}, \quad (4)$$

where W_{ℓ}^{VV} acts within V , W_{ℓ}^{UU} acts within U , and W_{ℓ}^{UV} captures cross-subspace interactions.

Let $X_{\ell} = V_{\ell} + U_{\ell}$, where $V_{\ell} \in V$ and $U_{\ell} \in U$. The evolution of the task-irrelevant component becomes

$$U_{\ell+1} = (I + W_{\ell}^{UU})U_{\ell} + W_{\ell}^{UV}V_{\ell}. \quad (5)$$

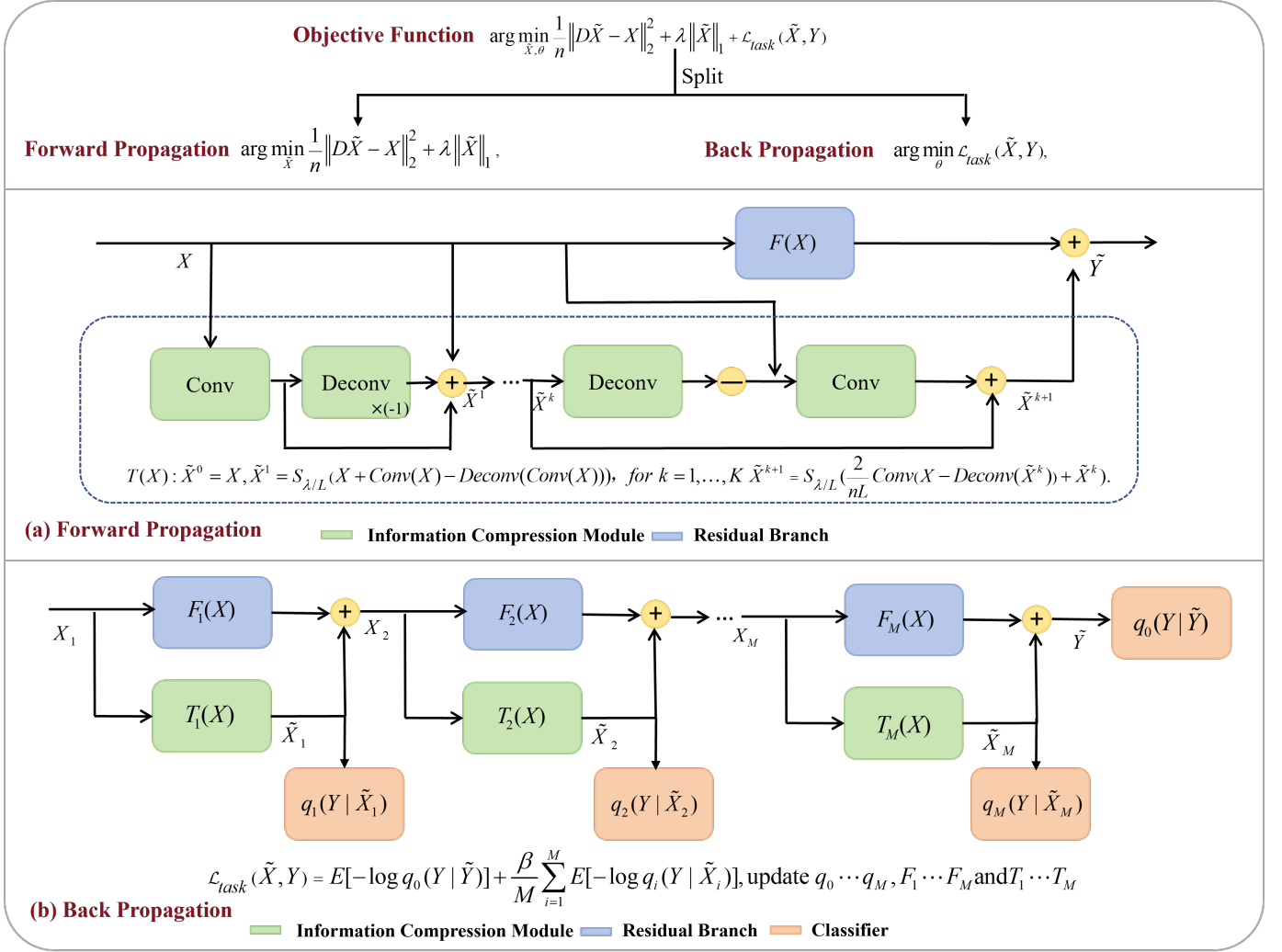


Fig. 2: The framework of the proposed adaptive compressed residual.

Compared with its non-residual counterpart,

$$U_{\ell+1}^{(NR)} = W_{\ell}^{UU} U_{\ell}^{(NR)} + W_{\ell}^{UV} V_{\ell}^{(NR)}, \quad (6)$$

the key difference lies in the identity term on U_{ℓ} . Even when W_{ℓ}^{UU} is contractive, the residual formulation always preserves U_{ℓ} through the shortcut connection. Moreover, cross-subspace interactions $W_{\ell}^{UV} V_{\ell}$ may continuously introduce additional task-irrelevant components, which are subsequently retained by later residual additions.

Therefore, under the stated linearized assumptions (i.e., W_{ℓ}^{UU} does not completely suppress U_{ℓ} and $W_{\ell}^{UV} \neq 0$), residual connections may preserve or accumulate task-irrelevant components more strongly than their non-residual counterparts. This observation provides a possible conceptual explanation for the empirical redundancy observed in deep residual models. This analysis is intended to provide qualitative motivation rather than a general proof for nonlinear CNN, Transformer, or Mamba architectures. In practical networks, nonlinear activations, normalization, attention, and input-dependent transformations may alter the behavior predicted by the linearized model.

Motivated by this observation, we propose to explicitly regulate the information propagated through the skip pathway. The proposed ACR module addresses this issue by selectively compressing residual features before fusion.

B. Information-Compressed Residual Connections

To address information redundancy in residual connections, we propose a residual design that selectively compresses the input feature, with the aim of preserving task-relevant information while suppressing redundant components.

Let the input to the ℓ -th compression module be

$$\mathcal{X}_{\ell} \in \mathbb{R}^{B \times C_{\ell} \times H_{\ell} \times W_{\ell}}, \quad (7)$$

where B , C_{ℓ} , H_{ℓ} , and W_{ℓ} denote the batch size, number of channels, height, and width, respectively. The learnable compression mapping is defined as

$$T_{\ell} : \mathbb{R}^{B \times C_{\ell} \times H_{\ell} \times W_{\ell}} \rightarrow \mathbb{R}^{B \times C_{\ell} \times H_{\ell} \times W_{\ell}}. \quad (8)$$

For notational simplicity, we omit the block index and use X and $\tilde{X}$ to denote the input and compressed output, respectively:

$$\tilde{X} = T(X), \quad (9)$$

where

$$X, \tilde{X} \in \mathbb{R}^{B \times C_\ell \times H_\ell \times W_\ell}. \quad (10)$$

Since the desired task-relevant representation is not directly observable, we use the ground-truth labels Y to guide the compressed representation $\tilde{X}$ toward preserving task-relevant information. Meanwhile, $\tilde{X}$ is required to retain sufficient information to reconstruct the input feature, i.e., $X \approx D\tilde{X}$, while remaining sparse. We obtain the following joint design objective:

$$\arg \min_{\tilde{X}, \theta} \frac{1}{2n} \|D\tilde{X} - X\|_2^2 + \lambda \|\tilde{X}\|_1 + \mathcal{L}_{\text{task}}(\tilde{X}, Y). \quad (11)$$

In order to facilitate the solution, Eq. (11) is decomposed into two subproblems:

$$\arg \min_{\tilde{X}} \frac{1}{2n} \|D\tilde{X} - X\|_2^2 + \lambda \|\tilde{X}\|_1, \quad (12)$$

$$\arg \min_{\theta} \mathcal{L}_{\text{task}}(\tilde{X}, Y). \quad (13)$$

Here, n is the number of samples contained in X , D is a learned dictionary matrix, and θ denotes the collection of trainable parameters.

The downstream loss $\mathcal{L}_{\text{task}}$ is detailed in Section III-D, and the regularization parameter λ controls the sparsity strength. The reconstruction term encourages $\tilde{X}$ to remain representative of X , the ℓ_1 penalty promotes a compact representation, and the task loss aligns $\tilde{X}$ with the downstream labels Y .

In the actual implementation, Eq. (12) motivates the unfolded compression module used in the forward pass. For each mini-batch, the network first performs a fixed number of K unfolded iterations to obtain $\tilde{X}$. The task loss in Eq. (13) is then evaluated, and a single backward pass jointly updates θ . Therefore, the practical training procedure is a jointly trained unfolded approximation of Eq. (11), rather than an exact alternating-optimization algorithm.

Finally, we integrate this compression module into the residual pathway by replacing the additive residual connection in Eq. (1) with the compressed residual (CR):

$$\tilde{Y} = \tilde{X} + F(X), \quad (14)$$

where $\tilde{X}$ is obtained by

approximately solving Eq. (12) through the fixed-depth unfolded module described in Section III-C.

In this way, the shortcut representation is encouraged to preserve task-relevant information while suppressing redundant components before residual fusion, thereby improving the compactness of deep feature representations.

C. Information Compression Module

Building upon the compressed residual formulation in Eq. (12), we now describe how to implement the solution efficiently within a trainable neural block. To solve Eq. (12) in an end-to-end manner, we employ the Iterative Shrinkage-Thresholding Algorithm (ISTA) [19], [32], which offers a

principled approach for solving ℓ_1 -regularized sparse coding problems. The iterative update for input X is given by:

$$\begin{aligned} \tilde{X} &= f(X) = \arg \min_{\tilde{X}} \frac{1}{2n} \|D\tilde{X} - X\|_2^2 + \lambda \|\tilde{X}\|_1 \\ &\approx \tilde{X}^{k+1} = S_{\lambda/L} \left(\frac{1}{nL} D^T (X - D\tilde{X}^k) + \tilde{X}^k \right), \end{aligned} \quad (15)$$

where L denotes the Lipschitz constant of the gradient ∇f , and S_r is the element-wise soft-thresholding operator defined as:

$$S_r(x) = \text{sign}(x) \cdot \max(|x| - r, 0). \quad (16)$$

To make the unfolded module compatible with residual architectures, we initialize the iterative variable as $\tilde{X}^{(0)} = X$. This initialization aligns with the identity-preserving property of residual connections, ensuring that the unfolded process begins from the same feature space as the skip input. Starting from X allows the reconstruction to remain close to the input, effectively retaining the essential and task-relevant components of the residual while adaptively suppressing redundancy through subsequent shrinkage steps [33].

To integrate ISTA into standard CNN, Transformer or Mamba architectures, we replace matrix multiplications with convolutional operations, interpreting $D^\top$ as a convolution and D as the corresponding transpose (deconvolution). The unfolded update is thus expressed as:

$$\begin{aligned} T(X) : \tilde{X}^0 &= X, \text{ for } k = 0, \dots, K \\ \tilde{X}^{k+1} &= S_{\lambda/L} \left(\frac{1}{nL} \text{Conv}(X - \text{Deconv}(\tilde{X}^k)) + \tilde{X}^k \right), \end{aligned} \quad (17)$$

where each iteration corresponds to one learnable layer consisting of a convolution, a deconvolution, and an element-wise shrinkage. This unfolded, fixed-depth architecture can therefore be implemented as a stack of K learnable layers, each performing convolution, deconvolution, and soft-thresholding operations, as illustrated in Fig. 2. At each residual block, the compressed reconstruction $\tilde{X}$ is obtained via the K -step unfolding and subsequently used to form the compressed residual connection:

$$\tilde{Y} = F(X) + T(X). \quad (18)$$

The dictionary operators and the block-specific threshold are shared across the unfolded ISTA steps within the same information compression module, but are not shared across different residual blocks or network stages. By incorporating the ISTA-based compression with identity-consistent initialization, the proposed module adaptively filters task-irrelevant activations and preserves essential residual information, leading to more compact and discriminative feature representations.

D. Cross-Entropy Guided Information Compression

The sparse reconstruction objective encourages the compressed representation to preserve information from the input feature, but it does not explicitly determine which components are relevant to the downstream task. We therefore introduce auxiliary classification supervision to guide each compressed residual representation toward label-discriminative information.

Let $\tilde{X}_i$ denote the compressed representation produced by the i -th CR module, where $i = 1, \dots, M$, and let $q_i(Y | \tilde{X}_i)$ denote the class-posterior distribution predicted by its auxiliary classifier. Here, M is the number of CR modules. Each auxiliary classifier first flattens $\tilde{X}_i$ into a feature vector and then applies a single-hidden-layer MLP. The auxiliary classifiers are jointly trained with the backbone but are removed during inference.

The connection between the auxiliary cross-entropy loss and mutual information can be established using a variational lower bound [34]. Let $p(Y | \tilde{X}_i)$ denote the unknown true posterior and let $q_i(Y | \tilde{X}_i)$ be its variational approximation. We have

$$\begin{aligned} I(\tilde{X}_i; Y) &= H(Y) + \mathbb{E}_{p(\tilde{X}_i, Y)} [\log p(Y | \tilde{X}_i)] \\ &= H(Y) + \mathbb{E}_{p(\tilde{X}_i, Y)} [\log q_i(Y | \tilde{X}_i)] \\ &\quad + \mathbb{E}_{p(\tilde{X}_i)} [D_{\text{KL}}(p(Y | \tilde{X}_i) \| q_i(Y | \tilde{X}_i))] \quad (19) \\ &\geq H(Y) + \mathbb{E}_{p(\tilde{X}_i, Y)} [\log q_i(Y | \tilde{X}_i)]. \end{aligned}$$

The inequality follows from the non-negativity of the KL divergence. Since $H(Y)$ is constant for a fixed dataset, minimizing

$$\mathcal{L}_{\text{aux}}^{(i)} = \mathbb{E} [-\log q_i(Y | \tilde{X}_i)] \quad (20)$$

maximizes a variational lower bound on $I(\tilde{X}_i; Y)$. This auxiliary objective encourages the compressed representation to preserve label-predictive information; the reconstruction and sparsity terms introduced previously provide the complementary compression constraints.

Let $\tilde{Y}$ denote the final representation produced by the ACR-enhanced backbone, and let $q_0(Y | \tilde{Y})$ denote the posterior predicted by the main classifier. The task loss is defined as

$$\mathcal{L}_{\text{task}} = \mathbb{E} [-\log q_0(Y | \tilde{Y})] + \frac{\beta}{M} \sum_{i=1}^M \mathbb{E} [-\log q_i(Y | \tilde{X}_i)]. \quad (21)$$

where $\beta = 0.1$ controls the contribution of the auxiliary supervision. The first term in Eq. (21) is the main classification loss, whereas the second term directly supplies task-guided gradients to the intermediate compressed representations.

E. Adaptive Compression-Residual Connections

In practice, exact reconstruction of the input feature is generally not achievable. We therefore write

$$X = D\tilde{X} + \varepsilon, \quad (22)$$

where ε denotes the reconstruction residual. The regularization parameter λ controls the trade-off between reconstruction fidelity and sparsity. In classical sparse estimation, an appropriate value of λ depends on the scale of the reconstruction residual, as summarized in the following standard result.

Lemma 1 [35]. Consider the linear model in Eq. (22), where $\varepsilon \sim \mathcal{N}(0, \sigma^2 I)$ is sub-Gaussian. Assume that the dictionary D satisfies the restricted eigenvalue condition over the support set

$\mathcal{S} = \text{supp}(\tilde{X})$, with $|\mathcal{S}| = s$, and let p denote the dimension of $\tilde{X}$, equivalently the number of dictionary atoms. If the regularization parameter is selected as

$$\lambda \asymp \sigma \sqrt{\frac{\log p}{n}}, \quad (23)$$

then the estimator $\hat{X}$ of $\tilde{X}$ satisfies

$$\|\hat{X} - \tilde{X}\|_2^2 = O_p\left(\frac{s\sigma^2 \log p}{n}\right). \quad (24)$$

Lemma 1 indicates that the theoretically appropriate regularization strength depends on the residual statistics. However, intermediate features at different network depths may exhibit different distributions and reconstruction characteristics. Consequently, using a manually selected and globally fixed value of λ may impose the same compression strength on residual blocks with different feature statistics.

Motivated by this observation, ACR directly treats λ as a trainable regularization parameter. Let λ_i denote the parameter associated with the i -th ACR block. In the unfolded update in Eq. (15), the fixed regularization parameter is replaced by λ_i , and the corresponding soft-thresholding level becomes λ_i/L_i , where L_i is the Lipschitz constant of the smooth reconstruction term in the i -th block. In our implementation, each ACR block learns its own λ_i , which is shared across the K unfolded iterations within that block. Parameters belonging to different ACR blocks are optimized independently, allowing the compression strength to vary across network depths. Each λ_i is initialized with a positive value and jointly optimized with the backbone, dictionary operators, and classifiers using the task objective in Eq. (21). During inference, the learned values remain fixed, and no additional online parameter estimation is required.

Therefore, ACR replaces the manually specified regularization coefficient in CR with a block-specific parameter learned from the training data and downstream supervision. For the theoretical analysis below, we assume that the learned regularization parameter satisfies the scaling condition in Eq. (23). Under this condition, the estimation behavior of the adaptive formulation is characterized by the following theorem.

Theorem (Estimation Error Retention) *If the estimator $\hat{\sigma}$ is consistent and $\hat{\sigma} - \sigma = \mathcal{O}_p(n^{-1/2})$, and if the regularization parameter λ satisfies*

$$\lambda \asymp \hat{\sigma} \sqrt{\frac{\log p}{n}}, \quad (25)$$

then the estimation error bound holds:

$$\|\hat{X} - \tilde{X}\|_2^2 = \mathcal{O}_P\left(\frac{s\sigma^2 \log p}{n}\right). \quad (26)$$

Proof. Define the error vector $\delta = \hat{X} - \tilde{X}$. Its optimality condition is

$$\frac{1}{n} D^\top (D\hat{X} - X) + \lambda z = 0, \quad z \in \partial \|\hat{X}\|_1. \quad (27)$$

Substituting Eq. (22) into Eq. (27) yields

$$\frac{1}{n} D^\top D\delta = \frac{1}{n} D^\top \varepsilon - \lambda z. \quad (28)$$

Using the maximal inequality for sub-Gaussian noise, we have

$$\left\| \frac{1}{n} D^\top \varepsilon \right\|_\infty = \mathcal{O}_P \left(\sigma \sqrt{\frac{\log p}{n}} \right). \quad (29)$$

Combining the consistency of $\hat{\sigma}$, i.e. $\hat{\sigma} - \sigma = \mathcal{O}_P(n^{-1/2})$, we obtain

$$\lambda = \mathcal{O}_P \left(\sigma \sqrt{\frac{\log p}{n}} \right). \quad (30)$$

Decompose δ into its support set S and its complement S^c :

$$\delta = \delta_S + \delta_{S^c}. \quad (31)$$

From the KKT condition, we can derive

$$\|\delta_{S^c}\|_1 \leq 3\|\delta_S\|_1. \quad (32)$$

Under the RE condition, for δ satisfying Eq. (32),

$$\frac{1}{n} \|D\delta\|_2^2 \geq \kappa \|\delta_S\|_2^2, \quad (33)$$

where $\kappa > 0$ is the RE constant.

Combining Eq. (28), (29)–(33), we obtain

$$\|\delta\|_2^2 = \mathcal{O}_P \left(\frac{s\sigma^2 \log p}{n} \right), \quad (34)$$

which completes the proof of Eq. (26). $\square$ Therefore, the estimation-error bound holds under the stated assumptions and the prescribed scaling condition on the regularization parameter. However, this conditional result does not imply that end-to-end training necessarily guarantees the same bound for arbitrary nonlinear network architectures.

IV. EXPERIMENTS

Experiments on image, video, and audio classification, together with object detection and semantic segmentation, demonstrate the effectiveness of ACR across diverse multimedia tasks and backbone architectures.

A. Experimental Setup

1) *Dataset*: We evaluate the proposed ACR on image, audio, and video classification benchmarks.

Image classification. We conduct experiments on ImageNet [36], CIFAR-10, and Mini-ImageNet [37]. ImageNet contains 1.2M training images and 50K validation images from 1,000 classes and serves as a large-scale benchmark. CIFAR-10 consists of 60K images (32×32) across 10 classes for small-scale evaluation. Mini-ImageNet, a 100-class subset of ImageNet, is used to assess generalization under limited data settings.

Audio classification. For audio understanding, we use ESC-50 [38], which includes 2,000 environmental audio clips evenly distributed across 50 categories with five-fold cross-validation splits.

Video classification. For temporal modeling evaluation, we adopt Something-Something v2 (SSv2) [39], a large-scale benchmark containing over 220K short video clips annotated with 174 human–object interaction categories.

These datasets collectively cover diverse modalities and scales, enabling comprehensive evaluation of ACR across heterogeneous multimedia tasks.

2) *Implementation Details*: Except for the ImageNet classification experiments, all reimplemented baselines and their ACR-equipped variants are trained from scratch on the same machine equipped with an Intel(R) Xeon(R) Platinum 8468V CPU, 100 GiB RAM, and a single NVIDIA A100-SXM4-40GB GPU. The ImageNet classification experiments are conducted using four NVIDIA A100 GPUs. All experiments are implemented using Python 3.10.19 and PyTorch 2.5.1 with CUDA 12.4 and cuDNN 9.0.1 on Ubuntu 20.04.6 LTS with NVIDIA driver version 550.54.14. Unless otherwise specified, ACR is inserted only into the last block of each backbone, and the unfolding depth is fixed to $K = 1$ in all experiments. For every baseline–ACR pair, we use identical training settings, including, but not limited to, the optimization strategy, training schedule, and data-processing pipeline. These model-specific settings follow the original implementations reported in the corresponding papers. No additional training tricks or optimization techniques are introduced for the ACR-equipped models. The only exception is Mini-ImageNet, for which we adopt the data augmentation and training protocol originally designed for ImageNet in the corresponding baseline models.

B. Image Classification

1) *ImageNet*: To evaluate the effect of ACR, we conduct experiments on ImageNet. Each baseline architecture and its ACR-enhanced counterpart are trained under identical optimization settings to ensure fair comparison. We adopt ResNet as a representative residual architecture, following the training protocol of SENet. Table I summarizes the results on ResNet-18/34/50, including variants with the final residual connection removed and those equipped with ACR. To quantify redundancy in the residual pathway, we measure the information bottleneck term $I(X; \tilde{X}) - I(\tilde{X}; Y)$. A lower value of this estimated information-bottleneck objective indicates a more favorable trade-off between input-information compression and label-information preservation under the adopted estimation protocol. The reported values are obtained by averaging the mutual-information estimates over the complete validation set. Natural logarithms are used in the estimation, and all reported mutual-information estimates and the resulting information-bottleneck scores are expressed in nats. For each validation sample, we extract the input X and the output $\tilde{X}$ of the final residual block, and the corresponding class label Y . The feature tensors are first spatially aggregated using global average pooling and are then standardized and reduced in dimensionality using PCA. We estimate $I(X; \tilde{X})$ from the paired real-valued feature representations using a k -nearest-neighbor mutual-information estimator. We similarly estimate $I(\tilde{X}; Y)$ using the corresponding nearest-neighbor estimator for real-valued features and discrete labels. All models use the same validation samples, preprocessing procedure, and estimator settings.

Across all model depths, removing the residual connection in the last block slightly decreases the information bottleneck score, confirming that unfiltered residual additions indeed introduce additional redundancy. However, this modification

TABLE I: Performance comparison between ResNet variants and ACR-enhanced models on ImageNet.

Model	$I(X; \tilde{X}) - I(\tilde{X}; Y)$	Prec@1(%)	Prec@5(%)	Params(M)	FLOPs(G)	Latency (ms/image)	Train Time (s/epoch)
ResNet-18	-0.234	70.56	89.56	11.69	1.819	0.070	348
-Residual Connection	-0.249	70.31	89.56	11.69	1.819	0.070	348
+ACR	-0.279	71.30	90.07	14.05	2.160	0.078	365
ResNet-34	-0.290	74.01	91.68	21.80	3.679	0.115	539
-Residual Connection	-0.310	73.26	91.22	21.80	3.679	0.115	539
+ACR	-0.372	74.64	92.03	24.16	4.018	0.118	554
ResNet-50	-0.338	76.18	93.00	25.56	4.134	0.222	976
-Residual Connection	-0.362	75.66	92.56	25.56	4.134	0.222	976
+ACR	-0.491	76.52	93.10	27.92	4.447	0.334	996

also leads to a consistent drop in classification accuracy, suggesting that simply discarding residual pathways harms representational capacity and cannot serve as an effective solution.

In contrast, integrating ACR yields both a substantially lower redundancy level and improved accuracy. For example, ACR reduces the information bottleneck score from -0.234 to -0.279 on ResNet-18 while increasing Top-1 accuracy to 71.30%. Similar improvements are observed on ResNet-34 and ResNet-50. These results indicate that ACR effectively suppresses redundant activations propagated by the skip connection while preserving and enhancing task-relevant information. Overall, the empirical evidence strongly supports our theoretical analysis: redundancy accumulates through residual connections in deep networks, and the proposed ACR provides a principled and effective mechanism to eliminate redundant information while retaining the benefits of residual learning.

2) *Mini-ImageNet*: To further validate the effectiveness of the proposed ACR mechanism, we integrate it into representative models based on different architectural paradigms, including the Transformer-based EfficientViT [40] and the Mamba-based Vision Mamba (Vim) [41], and evaluate them on the Mini-ImageNet image classification benchmark. As reported in Table II, we compare ACR with the original residual baselines and CBAM, a representative lightweight attention module. To determine whether the improvements of ACR are merely attributable to its additional parameters or computation, we further introduce parameter/FLOP-matched control modules. To ensure a controlled comparison, CBAM, ACR, MLP and Conv are inserted at the same location, namely, the residual pathway of the final block of each backbone. Moreover, all variants derived from the same backbone are trained using exactly the same optimization strategy, training schedule, data augmentation, and other hyperparameters. Under these controlled settings, ACR consistently achieves the highest Top-1 accuracy across all four backbones. Compared with CBAM, ACR improves Top-1 accuracy by 4.72, 0.81, 0.38, and 0.17 percentage points on Vim-Ti, Vim-S, EfficientViT-M0, and EfficientViT-M1, respectively. Compared with the parameter/FLOP-matched controls, ACR further improves Top-1 accuracy by 1.52, 0.66, 0.06, and 0.71 percentage points, respectively. These results indicate that the gains of ACR cannot be explained solely by additional model capacity or computation. Instead, they support the effectiveness of the proposed task-guided compression mechanism for improving feature representation across heterogeneous architectures.

3) *Visualization Analysis*: To provide a more intuitive demonstration of the effectiveness of ACR, we employ Grad-CAM to visualize the attention regions of the models. Specifically, we compare the heatmaps generated by EfficientViT and EfficientViT-ACR. As shown in Fig. 3, EfficientViT equipped with ACR focuses more accurately on the target object regions. Moreover, the responses produced by EfficientViT-ACR exhibit smaller activation areas and stronger response intensities, indicating that ACR encourages the network to concentrate on more discriminative and compact regions. This can be attributed to the sparsity caused by combining with ACR, which enables the model to learn compact feature representations.

To further examine the evolution of feature responses across network depths, we visualize the Grad-CAM heatmaps from three successive stages of EfficientViT-ACR in Fig. 4. The earlier stages exhibit relatively dispersed responses covering both the target objects and surrounding backgrounds. As the features propagate to deeper stages, the high-response regions become progressively more concentrated on discriminative object regions. In particular, the final stage containing the ACR-equipped block preserves prominent object responses while suppressing background activations. Together with the direct baseline comparison in Fig. 3, these layer-wise visualizations provide qualitative evidence that ACR promotes more compact and spatially selective deep representations.

We further investigate the impact of ACR from a global feature-space perspective. Fig. 5 (a) shows a per-class margin heatmap on Mini-ImageNet, where each cell encodes the average pairwise distance between two classes. We map smaller margins to cool hues (blues) and progressively larger margins to warm hues (yellows)—the yellower a cell, the greater its inter-class separation. In the baseline model (left), most classes lie in the cooler-to-neutral range, indicating relatively modest separation. After integrating our ACR (right), the heatmap shifts noticeably toward warm tones: a majority of classes now display bright yellow cells, signifying substantially enlarged margins. This shift quantitatively confirms that ACR improves the discriminability of features by pushing class centroids farther apart. The t-SNE visualizations in Fig. 5 (b) reveal that models equipped with ACR exhibit significantly reduced intra-class variance, with samples forming denser and more cohesive clusters. Meanwhile, inter-class regions become more separated, indicating improved discriminability.

TABLE II: Comparison results of Vision Mamba and EfficientViT models.

Model	Prec@1 (%)	Prec@5 (%)	Params (M)	FLOPs (G)	Latency (ms/image)	Train Time (s/epoch)
Vim-Ti	62.62	85.41	6.97	1.343	12.859	113
+CBAM	65.65	86.74	6.98	1.343	12.901	113
+MLP	68.85	89.68	6.98	1.358	12.845	114
+ACR	70.37	90.30	6.98	1.358	12.904	114
Vim-S	74.61	90.97	25.45	4.778	13.173	243
+CBAM	76.75	92.26	25.47	4.778	13.237	245
+MLP	76.90	92.85	25.51	4.836	13.007	243
+ACR	77.56	92.69	25.51	4.836	13.223	244
EfficientViT-M0	74.96	91.79	2.349	0.159	7.683	31
+CBAM	74.68	92.08	2.354	0.159	7.832	32
+Conv	75.00	92.22	2.682	0.180	8.110	32
+ACR	75.06	91.97	2.681	0.180	8.189	32
EfficientViT-M1	79.22	93.85	2.981	0.337	6.485	31
+CBAM	79.44	93.93	2.985	0.337	7.412	31
+Conv	78.90	93.89	3.313	0.358	7.161	31
+ACR	79.61	93.96	3.312	0.358	7.289	32

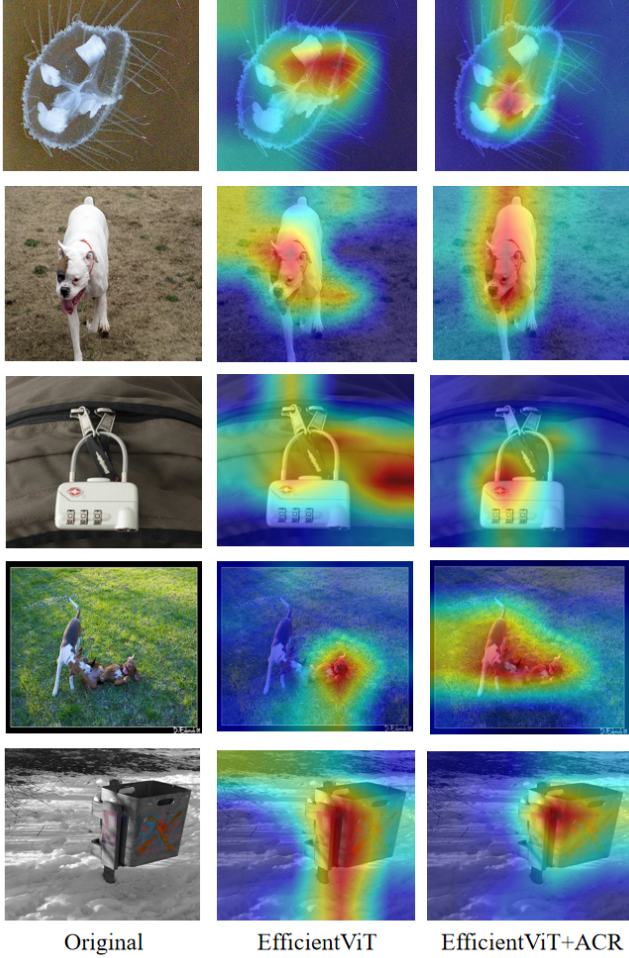


Fig. 3: Heatmap comparisons of EfficientViT and EfficientViT+ACR on the Mini-ImageNet dataset. Red indicates strong activation for the predicted class; blue indicates weak activation.

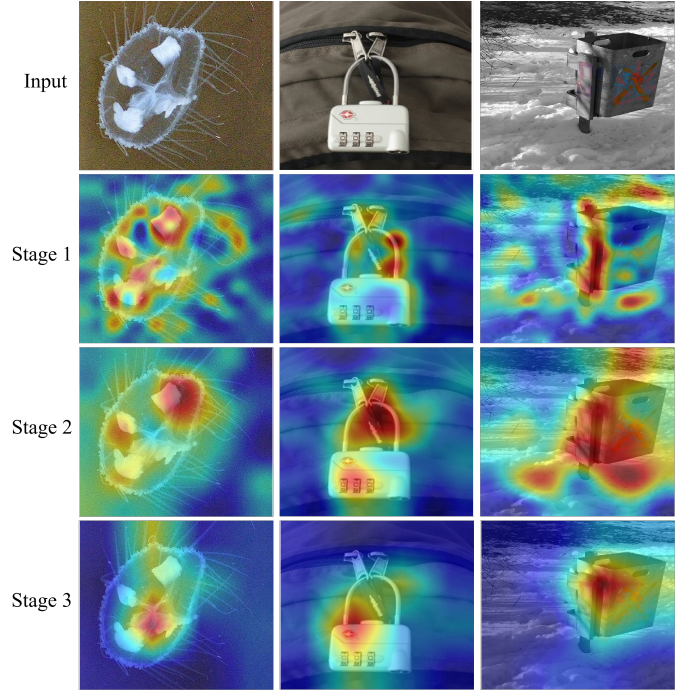


Fig. 4: Stage-wise Grad-CAM visualizations of EfficientViT-ACR.

C. Classification Results on Additional Media

1) *Audio*: To further evaluate the multimedia effectiveness of the proposed ACR mechanism in the audio domain, we integrate it into two representative audio models, namely PANN [42] and HTS-AT [43], and conduct audio classification experiments on the ESC-50 benchmark. As shown in Table III, under identical training and evaluation settings, we compare the original baseline models with their ACR-enhanced counterparts. The experimental results demonstrate that the introduction of ACR consistently improves classification accuracy on ESC-50, indicating that ACR can effectively enhance audio feature representation and model optimization, and generalizes well to audio classification tasks.

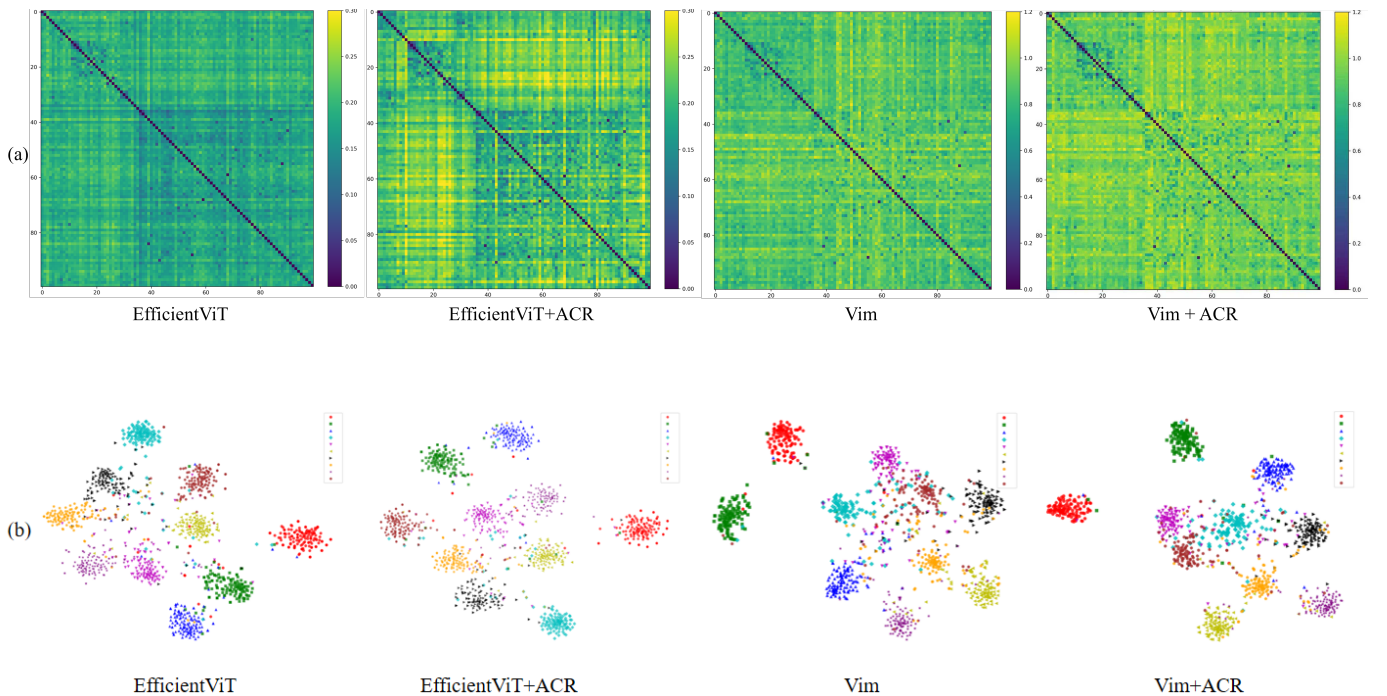


Fig. 5: Global feature-space analysis of EfficientViT, EfficientViT + ACR, Vim and Vim + ACR on Mini-ImageNet. (a) Class-distance heatmaps. (Brighter yellow regions indicate larger inter-class distances.) (b) t-SNE visualizations.

To better understand the effect of the proposed ACR module, we visualize the feature maps of the first five attention heads in the final block of both HTS-AT and HTS-AT equipped with ACR. The comparison in Fig. 6 shows that, after introducing ACR, the activation patterns become noticeably sparser. This increased sparsity indicates that ACR enhances feature selectivity and promotes more efficient representation learning, which helps explain the observed performance improvements.

TABLE III: Comparison results of audio classification models.

Model	Prec@1(%)	Params(M)	FLOPs(G)	Latency (ms/audio)	Train Time (s/epoch)
HTS-AT	97.0	28.9	7.024	5.610	35
+ACR	97.5	30.0	7.175	6.028	35
PANN	95.5	4.7	1.442	3.276	18
+ACR	95.8	5.2	1.499	4.184	19

2) *Video*: To further verify the multimedia effectiveness of the proposed ACR mechanism, we integrate it into representative video models built upon different architectural paradigms, including the Transformer-based UniFormer [44] and the Mamba-based VideoMamba [45], and evaluate their performance on the SSv2 video classification benchmark in Table IV. By conducting controlled comparisons between the original baseline models and their ACR-enhanced variants under identical training settings, the experimental results show that incorporating ACR consistently improves classification accuracy on SSv2. These results demonstrate that ACR effectively enhances temporal feature modeling and representation learning across heterogeneous video architectures, confirming its strong applicability to video classification tasks.

D. Semantic Segmentation

To evaluate whether ACR generalizes beyond classification, we integrate it into Mask2Former with Swin-T and Swin-S backbones and conduct semantic segmentation experiments on Cityscapes. As shown in Table V, ACR consistently improves the segmentation performance of both backbones. For Mask2Former-Swin-T, ACR increases mIoU from 78.26% to 78.70% and mAcc from 87.38% to 87.89%, yielding gains of 0.44 and 0.51 percentage points, respectively. For the stronger Mask2Former-Swin-S baseline, ACR improves mIoU by 1.28 percentage points, from 80.65% to 81.93%, while increasing mAcc from 89.71% to 90.13%. These improvements introduce only modest additional complexity: ACR adds approximately 0.59M parameters and 2.42G FLOPs to each model, while the inference latency increases by less than 1 ms per image. The training times also remain comparable to those of the corresponding baselines. These results indicate that ACR can improve dense pixel-level prediction without substantially altering the computational profile of Mask2Former.

E. Object Detection

We further evaluate ACR on object detection by integrating it into RT-DETR with ResNet-18 and ResNet-34 backbones and conducting experiments on the PASCAL VOC 2012 dataset. As reported in Table VI, ACR provides consistent improvements across both detector configurations. For RT-DETR-R18, ACR improves AP from 57.137% to 57.410%, with corresponding improvements in AP₅₀ and AP₇₅. For RT-DETR-R34, ACR increases AP from 60.234% to 60.498% and produces a more pronounced improvement in AP₇₅, from 64.532% to 65.407%. The additional model size is limited to

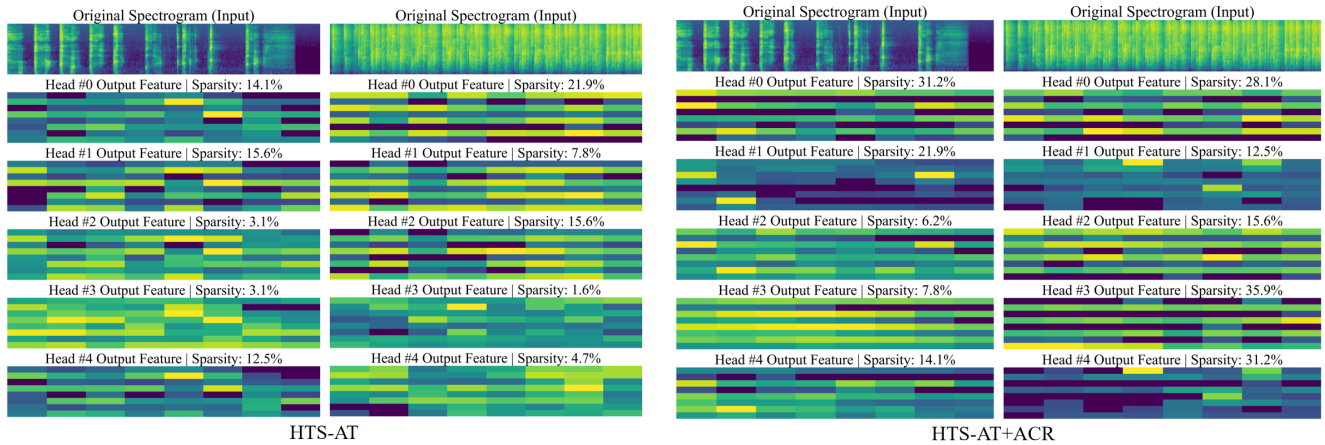


Fig. 6: Visualization of the feature maps of the first five heads from the final block. Colors indicate activation magnitude, where darker blue regions correspond to lower feature intensity, indicating a sparser representation.

TABLE IV: Comparison between baseline models and their ACR-augmented versions on the SSv2 dataset.

Model	Prec@1(%)	Prec@5(%)	Params (M)	FLOPs (G)	Latency (ms/clip)	Train Time (s/epoch)
UniFormer-S	67.70	91.44	21.28	41.757	16.269	8925
+ACR	68.47	91.74	22.07	42.373	16.726	9706
VideoMamba-T	63.15	87.62	6.99	16.889	12.220	3821
+ACR	63.54	87.82	7.03	17.005	13.475	3832
VideoMamba-S	63.97	88.61	25.48	33.778	13.478	7236
+ACR	64.54	88.65	25.63	34.241	13.936	7447

TABLE V: Performance comparison between Mask2Former baselines and ACR-enhanced models on the Cityscapes dataset.

Model	mIoU (%)	aAcc (%)	mAcc (%)	Params (M)	FLOPs (G)	Latency (ms/image)	Train Time (min/5k iters)
Mask2Former-Swin-T	78.26	96.39	87.38	47.406	544.379	136.025	31.37
+ACR	78.70	96.42	87.89	47.996	546.795	136.653	31.50
Mask2Former-Swin-S	80.65	96.50	89.71	68.724	730.820	167.831	37.96
+ACR	81.93	96.64	90.13	69.314	733.235	168.828	37.95

TABLE VI: Performance comparison between RT-DETR baselines and ACR-enhanced models on the PASCAL VOC 2012 dataset.

Model	AP (%)	AP50 (%)	AP75 (%)	Params (M)	FLOPs (G)	Latency (ms/image)	Train Time (min/epoch)
RT-DETR-R18	57.137	75.879	61.450	20.107	30.225	7.152	5.74
+ACR	57.410	76.008	61.579	20.255	32.112	7.321	5.85
RT-DETR-R34	60.234	79.230	64.532	31.349	46.304	8.954	6.61
+ACR	60.498	79.394	65.407	31.497	48.191	9.020	6.64

0.148M parameters for both backbones. Although the unfolded compression module increases FLOPs, the measured latency increases by only 0.169 ms per image for RT-DETR-R18 and 0.066 ms per image for RT-DETR-R34 and the training time remains essentially unchanged. The consistent gains across AP thresholds suggest that ACR is applicable to object detection tasks while maintaining a modest practical overhead.

F. Ablation Study

1) *Learnable Compression Coefficient*: To further validate the effectiveness of the learnable compression coefficient, we evaluated our methods on the CIFAR-10 image classification benchmark using five ResNet variants (ResNet-20/32/44/56/110). As a sanity check, our re-implementation of the original residual networks achieves

marginally better test errors than reported. We then replace the vanilla addition in each residual block with our compressed residual (CR) and, in a second setup, with the adaptive compressed residual (ACR). As shown in Table VII, incorporating CR yields an average error reduction of 0.26% across all depths, while ACR further boosts performance. These results demonstrate that both CR and its adaptive variant consistently improve classification accuracy.

2) Location of Adaptive Compressed Residual Connections:

To investigate at which stage adaptive compressed residual connections are most effective, we replaced the standard residual blocks in each ResNet stage with our compressed residual connection and evaluated performance on CIFAR-10. As shown in Table VIII, inserting compressed connections in the final stage yields the largest accuracy gains, while early-stage

TABLE VII: Performance comparison between ResNet variants and ACR-enhanced models on CIFAR-10.

Model	Test error original(%)	Test error reimplementation(%)	Test error +CR(%)	Test error +ACR(%)
ResNet-20	8.75	8.27	7.89 (↓0.38)	7.79 (↓0.48)
ResNet-32	7.51	7.37	7.25 (↓0.12)	7.13 (↓0.24)
ResNet-44	7.17	7.25	6.99 (↓0.26)	6.87 (↓0.38)
ResNet-56	6.97	6.97	6.69 (↓0.28)	6.64 (↓0.33)
ResNet-110	6.43	6.32	6.07 (↓0.25)	5.92 (↓0.40)

TABLE VIII: Ablation study of ACR on different locations in ResNet-18.

Model	Prec@1(%)	Params(M)	FLOPs(G)
ResNet-18	91.89	11.69	1.819
-First stage	92.09	11.76	2.507
-All	92.28	16.24	4.055
-Last stage	92.46	15.10	2.332
-Last block	92.33	14.05	2.160

replacements offer only marginal improvement. This behavior arises because shallow layers prioritize fitting raw features, whereas deeper layers focus on compressing and filtering out task-irrelevant information; imposing compression too early disrupts the fitting process [46]. Moreover, task-irrelevant information accumulates as the network deepens, so applying our compression module in late stages is naturally more effective. Finally, we observed that replacing just the last residual block already delivers strong performance, striking a superior balance between accuracy, parameter count, and FLOPs.

V. CONCLUSION

In this work, we revisited the internal information dynamics of residual connections and challenged the long-held assumption that their additive form is always beneficial. Through theoretical analysis and empirical investigation, we identified that standard residual links propagate a considerable amount of redundant information, particularly in deeper layers where abstraction and compression should dominate. To examine and mitigate this redundancy, we proposed an Adaptive Compressed Residual (ACR) that integrates a learnable sparse-coding mechanism into the skip pathway. By reformulating feature fusion as a compression-reconstruction problem, ACR adaptively filters task-irrelevant activations while preserving discriminative content. Experimental results on multiple CNN- and Transformer-based architectures reveal that even moderate compression improves generalization and discriminability, validating our hypothesis about residual redundancy. This study contributes a new perspective on the internal efficiency of residual learning—showing that not all information preserved by residuals is beneficial, and that their redundancy can be both modeled and reduced. We believe this insight opens new directions for designing redundancy-aware architectures that better balance information preservation and compression in deep representation learning.

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